Likelihood targets of a nested ensemble Kalman filter

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Abstract

State-space models with a static parameter have an observed-data likelihood that integrates out a latent Markov process. Nested particle filters for the parameter become expensive as the observation dimension grows. Replacing each likelihood increment by an ensemble Kalman forecast changes the posterior, and replacing that forecast by a nearest-neighbour prediction changes it again. These are different errors. We separate them. The plain filter targets the ensemble Kalman posterior. The nearest-neighbour weight targets the surrogate posterior. A Rao-Blackwellised weight puts the true observation density back, whilst the forecast stays Gaussian. If the observation density is already that Gaussian density, the two weights agree. On a Lotka–Volterra diffusion, with prey observed under a Gaussian density whose variance equals the latent prey, we compare the plain filter, the Rao-Blackwellised filter and bootstrap sequential Monte Carlo squared. The plain filter lies 0.0586 and 0.0354 below the bootstrap filter on the log predation coefficient and the log predator death rate. Neither gap exceeds its standard error, so the shift is unresolved. No bound between the surrogate posterior and the ensemble Kalman posterior was obtained.

Keywords: Lotka–Volterra model, Rao-Blackwellisation, sequential Monte Carlo, state-space model, surrogate likelihood

1 Introduction

State-space models are used when a latent Markov process is observed with noise and a static parameter governs the transition and the observation. The observed-data likelihood is an integral over the latent path, and that integral is rarely available in closed form. Sequential Monte Carlo squared (SMC2) propagates a cloud of parameter particles, reweights them by a particle-filter estimate of each likelihood increment, and moves them with a particle Metropolis–Hastings kernel when the effective sample size falls (Chopin et al. 2012). The variance of that estimate grows with the length of the series and, more sharply, with the observation dimension.

The ensemble Kalman filter replaces the forecast distribution by a Gaussian distribution whose mean and covariance are the sample moments of an ensemble, and it updates the ensemble with a Kalman shift. The likelihood contribution is then the Gaussian marginal of that forecast. It is not unbiased for the true likelihood. Golightly et al. (2026) place this contribution inside the SMC2 recursion. The resulting nested ensemble Kalman filter targets the parameter posterior induced by the ensemble Kalman likelihood, which we denote $\hat{\pi}_{\mathrm{enkf}}$, rather than the exact posterior π_t.

For a nonlinear or non-Gaussian observation the same construction replaces the observation density by a linear Gaussian approximation before the ensemble update. The plain filter still targets $\hat{\pi}_{\mathrm{enkf}}$, now under that approximation. A Rao-Blackwellised weight puts the true observation density back into the increment, while the forecast remains Gaussian, and targets a different posterior, which the source paper denotes $\tilde{\pi}_{\mathrm{enkf}}$. Vanilla SMC2, with a bootstrap proposal, targets π_t. Separately, the nearest-neighbour surrogate that Golightly et al. (2026) use to screen a Metropolis–Hastings proposal can be used as the incremental weight itself. That choice

defines a further posterior, which we denote π_{knn}. A screen that is corrected by the ensemble Kalman likelihood leaves $\hat{\pi}_{\mathrm{enkf}}$ invariant. A weight that is the surrogate does not.

These are two errors, and a comparison that jumps from π_{knn} to π_t mixes them. The first error is the ensemble Kalman likelihood against the true likelihood, hence $\hat{\pi}_{\mathrm{enkf}}$ against π_t. The second is the nearest-neighbour prediction against the ensemble Kalman likelihood, hence π_{knn} against $\hat{\pi}_{\mathrm{enkf}}$. On the Lotka–Volterra diffusion in Section 5.2 of Golightly et al. (2026), the observation is Gaussian with variance equal to the latent prey state. That paper reports that the plain filter shifts the posterior means of θ_2 and θ_3 because the observation variance is replaced by a function of the forecast mean, and that the Rao-Blackwellised filter reduces the discrepancy at greater cost. The distance between π_{knn} and $\hat{\pi}_{\mathrm{enkf}}$ is not bounded there.

Our contribution is to separate the two errors and to compare the three targets on that nonlinear observation model. (i) The plain nested filter targets $\hat{\pi}_{\mathrm{enkf}}$. The nearest-neighbour incremental weight targets π_{knn}, and we do not treat those posteriors as the same object. (ii) When the true observation density equals the linear Gaussian density inside the filter, the Rao-Blackwellised weight equals the ensemble Kalman weight. The identity is specific to that case. (iii) On the Lotka–Volterra observation we compare the plain filter, the Rao-Blackwellised filter and bootstrap SMC2, under a reduced design that is stated as such. A bound on the distance between π_{knn} and $\hat{\pi}_{\mathrm{enkf}}$ was not obtained. The paper reports the diagnostic that was fixed before any output was seen. A faster neighbour search, a kernel chosen by time, and a finite-difference proposal were examined in an earlier pilot and are not contributions here.

The identity is checked on a scalar Ornstein–Uhlenbeck diffusion with a known Kalman likelihood, at a smaller particle count and a shorter series than Section 5.1 of Golightly et al. (2026). The nonlinear comparison uses one prey series from the Lotka–Volterra diffusion, with fewer parameter particles and fewer algorithm replicates than Section 5.2, and without the particle Markov chain Monte Carlo gold standard used there. No epidemic series is analysed. Temfack and Wyse (2026) already study a nested ensemble scheme for epidemic incidence.

The remainder of this paper is organised as follows. Section 2 describes the neighbouring algorithms and which posterior each one targets. Section 3 separates the two errors and records the identity. Section 4 reports the identity check and the Lotka–Volterra comparison, and Section 5 discusses what those runs support and what remains open.

2 Related work

SMC2 propagates a cloud of parameter particles, and each particle carries a particle filter. The incremental weight is an unbiased estimate of the likelihood contribution. Chopin et al. (2012, Proposition 1) write the extended target so that its θ-marginal is the exact parameter posterior for any fixed number of state particles. The rejuvenation kernel is particle Metropolis–Hastings on that extended target. Replacing the inner particle filter by an ensemble Kalman filter removes the unbiasedness on which this marginal property rests. The marginal is then $\hat{\pi}_{\mathrm{enkf}}$, not π_t.

Ensemble Markov chain Monte Carlo uses the same ensemble Kalman likelihood inside one Metropolis–Hastings chain on a fixed data set (Drovandi et al. 2022). One rejuvenation sweep of the nested filter is one iteration of that scheme, but the chain itself is not sequential. Even when the forecast is exactly Gaussian, the plug-in Gaussian density is a biased estimator of the idealised Gaussian predictive. An unbiased estimator of that Gaussian density, of the kind used by Drovandi et al. (2022) and by Temfack and Wyse (2026), does not restore the exact posterior of the state-space model. It removes a different error from the one that separates $\hat{\pi}_{\mathrm{enkf}}$ from π_t.

A different sequential construction gives the parameter an artificial transition and resamples it with a shrinkage kernel, so that the filter does not collapse onto a single parameter value (Liu and West 2001). The particle ensemble Kalman filter of Katzfuss et al. (2018) weights parameter particles by an integrated ensemble Kalman likelihood and, for a constant parameter, moves them with that kernel. The nested filter keeps θ static and restores diversity by a Metropolis–

Hastings move under $\hat{\pi}_{\text{enkf}}$.

Delayed acceptance screens a proposal with a cheap approximation and then corrects the screen with the intended target (Christen and Fox 2005). If the parent kernel is irreducible for that target, the proposal is reversible, and the approximation is positive wherever the proposal is positive, the target remains invariant. Sherlock et al. (2016) store the expensive evaluations in a KD-tree and use an inverse-distance neighbour average as the screen, inside batch Markov chain Monte Carlo. Bon et al. (2021) put a surrogate inside the mutation of a sequential Monte Carlo sampler for a static model. In both cases the second stage evaluates the expensive target. Sherlock et al. (2016, Section 5.1) contrast this with algorithms that target the approximate posterior itself. The nearest-neighbour weight in Section 3 is that second use. It targets π_{knn}. The screen, when it is corrected, does not.

Temfack and Wyse (2026) also replace the inner particle filter of SMC2 by an ensemble Kalman filter, for epidemic incidence, with a state-dependent observation variance. They are explicit that the Gaussian assumption remains. Their mutation is particle marginal Metropolis–Hastings. They do not use the Rao-Blackwellised weight of Section 3, and this paper does not reanalyse an epidemic series.

Two further modifications of the resample-move step are set aside. A kernel switch that depends on a conditional particle filter, as in Botha et al. (2023), is not available once the inner filter is an ensemble Kalman filter, and the unbiasedness used by Fearnhead and Taylor (2010) is not the ensemble Kalman weight. A finite-difference proposal for the gradient of the ensemble Kalman log-likelihood can be corrected by Metropolis–Hastings, in the linear Gaussian case covered by the score result of Chen et al. (2021, Theorem 3.2), but the pilot comparison did not support the proposal on cost. An uncorrected Langevin proposal (Rosato et al. 2024) is a derivative of a particle-filter likelihood and is not used.

3 Methods

The observation model in the linear Gaussian case is

$$f_t^{(\theta)}(y_t \mid x_t) = \mathcal{N}(y_t; Hx_t, R). \tag{1}$$

An ensemble of size N is propagated by the state transition. The forecast mean $\hat{\mu}_{t|t-1}$ and covariance $\hat{\Sigma}_{t|t-1}$ are the sample moments, with divisor $N-1$. The ensemble Kalman likelihood contribution is

$$\hat{p}^{(\theta)}_{\text{enkf},t}(y_t \mid y_{0:t-1}) = \mathcal{N}\big(y_t; H\hat{\mu}_{t|t-1},\, H\hat{\Sigma}_{t|t-1}H^\top + R\big). \tag{2}$$

This contribution is exact for the Gaussian forecast approximation and is not unbiased for the true predictive density. The product of these contributions, multiplied by the prior, defines $\hat{\pi}_{\text{enkf},t}$ up to the ensemble randomness carried by the extended target of Golightly et al. (2026, Section 4.1). The exact posterior at the same time is $\pi_t(\theta \mid y_{0:t})$. The first error is the difference between these two distributions.

3.1 Plain nested filter

The outer recursion follows Algorithm 6 of Golightly et al. (2026). At time t, each parameter particle is updated by one ensemble Kalman step and its weight is multiplied by (2). If the effective sample size is at least γM, the cloud is left as it is. Otherwise the particles are resampled with replacement, the weights are reset to $1/M$, and each particle is offered one Metropolis–Hastings proposal. The proposal is a random walk whose covariance is ζ times the sample covariance of the other resampled particles. In the studies below, $\zeta^2 = 2.56^2/d_\theta$ and the ensemble size is held fixed. The acceptance ratio uses the prior and the ensemble Kalman likelihood of the series so far. The likelihood at a proposed value is obtained by rerunning the filter from the known initial condition.

The printed Algorithm 6 places the effective-sample-size test inside the particle loop. The surrounding text applies one resample-move step when the effective sample size of the updated cloud falls below γM. We follow the surrounding text.

For the nonlinear Gaussian observation in Section 4, the matrix R in (2) is replaced by the scalar $\tilde{R} = H\hat{\mu}_{t|t-1}$, as in Section 5.2 of Golightly et al. (2026). The plain filter then targets $\hat{\pi}_{\mathrm{enkf}}$ under that choice. It does not evaluate the true observation density.

3.2 Rao-Blackwellised weight

For a general observation density f, the filter can still be run under a linear Gaussian approximation $g_t^{(\theta)}(y_t \mid x_t) = \mathcal{N}(y_t; \tilde{H}x_t, \tilde{R})$. The Rao-Blackwellised increment averages, over the analysis ensemble, the terms

$$\hat{g}^{(\theta)}_{\mathrm{enkf},t}(y_t)\,\frac{f_t^{(\theta)}(y_t \mid x_t^j)}{g_t^{(\theta)}(y_t \mid x_t^j)}. \tag{3}$$

The parameter target induced by these increments is $\tilde{\pi}_{\mathrm{enkf},t}$. The ratio f/g puts the true observation density back. The Gaussian forecast remains, so $\tilde{\pi}_{\mathrm{enkf},t}$ is not π_t.

Lemma 1. *If $f_t^{(\theta)} = g_t^{(\theta)}$, the Rao-Blackwellised increment equals the ensemble Kalman increment* (2).

The ratio in (3) is then equal to 1 for every analysis member, and the average of those terms is the predictive likelihood under g, which is (2). The incremental weights agree, and so do the parameter targets. Evaluating the ratio costs an extra density for each ensemble member and does not change the estimate. The cancellation uses equality of the two observation densities. It does not, by itself, describe a nonlinear observation.

In the Lotka–Volterra comparison the true density f is Gaussian with variance equal to the prey component. The plain filter uses $\tilde{R} = H\hat{\mu}_{t|t-1}$. The Rao-Blackwellised filter uses the same quantity multiplied by 2, which is the inflation stated in Section 5.2 of Golightly et al. (2026), and it evaluates f/g at the analysis members. Vanilla SMC2 does not use g. Its state weight is f, its proposal is the Euler–Maruyama transition, and its parameter target is π_t.

3.3 Nearest-neighbour weight

If the incremental weight is a nearest-neighbour prediction of (2), the weights target

$$\pi_{\mathrm{knn},t}(\theta) \propto \pi(\theta)\,\exp\bigl(\ell_{\mathrm{knn},t}(\theta)\bigr), \tag{4}$$

conditional on the stored cloud that defines the surrogate log-likelihood $\ell_{\mathrm{knn},t}$. This is the second error. Equation (4) is not $\hat{\pi}_{\mathrm{enkf},t}$, and a comparison of $\pi_{\mathrm{knn},t}$ with π_t mixes the two errors.

In the linear Gaussian pilot the first k particles are a reference set and are always updated with the ensemble Kalman filter. Every other particle takes the inverse-distance prediction of the reference increments. The diagnostic, fixed before that run, recomputes (2) for eight non-reference particles at the terminal time. The surrogate increment was to be called a poor approximation of the ensemble Kalman increment if the mean, over replicates, of the median absolute log difference exceeded 1.

A uniform bound δ on $|\ell_{\mathrm{knn},t}(\theta) - \ell_{\mathrm{enkf},t}(\theta)|$ over the prior support would imply that the total variation distance between the two posteriors built from those fixed log-likelihoods is at most $\min\bigl(1, (e^{2\delta}-1)/2\bigr)$. The short calculation is recorded in the theory note of 15 October 2026. The hypothesis was not proved. The ensemble Kalman log-likelihood is a random function of the ensemble, the diagnostic is not a uniform δ, and no Lipschitz constant for that log-likelihood was available from the papers we used. The diagnostic remains the comparison we report. We do not state a theorem for the distance.

The same pilot also compared an exact tree with a list, a kernel switch that depends only on time, and a finite-difference random-walk proposal. Those comparisons are not part of the claim. The finite-difference proposal leaves $\hat{\pi}_{\text{enkf}}$ invariant when the Gaussian proposal density is included in the acceptance ratio, and the pilot did not support it on effective sample size per transition draw.

4 Simulation studies

Two checks are reported. The first is the linear Gaussian identity, on the Ornstein–Uhlenbeck pilot already stored in the results files. The second is the Lotka–Volterra comparison of Section 5.2 of Golightly et al. (2026), at the reduced settings fixed before the run. All algorithms were coded in base R and the Lotka–Volterra runs used R 4.3.3 on an Intel Xeon processor. The pilot is not a reproduction of the 2026 tables.

4.1 Identity and the surrogate diagnostic

The identity check uses the scalar Ornstein–Uhlenbeck diffusion of Section 5.1 of Golightly et al. (2026), with $M = 60$ parameter particles, 16 observations, a fixed ensemble of size 12 and four replicate series. The gold-standard chain on each series has 3000 iterations after a burn-in of 1000. Those choices are smaller than Section 5.1. When the observation variance inside the filter equals the observation variance of the data, $R_g = R = 0.1$, the log ensemble Kalman increment and the log Rao-Blackwellised increment agree. The maximum absolute difference on each replicate is 0. Lemma 1 is the corresponding statement. The correction does not change the weight in this case, and the extra density evaluations have nothing to correct.

The same pilot records the surrogate diagnostic. The mean, over four replicates, of the median absolute difference between a nearest-neighbour increment and a fresh ensemble Kalman increment is 0.277. The threshold fixed before the run was 1. The diagnostic stays inside that threshold. It is not a uniform bound on the log-likelihood, and it is not substituted into the total-variation calculation of Section 3.

The pilot also compared an exact tree with a list and a time-based kernel switch with two fixed kernels, and it compared a finite-difference proposal with a random walk. Those runs are not part of the claim.

4.2 Lotka–Volterra targets

The latent process is the prey–predator diffusion (25) of Golightly et al. (2026), with drift

$$a(x, \theta) = \big(\theta_1 x_1 - \theta_2 x_1 x_2,\ \theta_2 x_1 x_2 - \theta_3 x_2\big)$$

and with the diffusion matrix printed in that section. The observation model is (27): the prey component is observed with a Gaussian density whose mean and variance both equal that component, and the predator component is latent. The truth is $x_0 = (50, 50)^\top$ and $\theta = (0.5, 0.0025, 0.3)^\top$. The prior is independent, $\theta_1 \sim \text{Gamma}(2, 4)$, $\theta_2 \sim \text{Gamma}(20, 10^4)$ and $\theta_3 \sim \text{Gamma}(2, 4)$, in the shape-rate parameterisation. The source paper writes $\text{Gamma}(a, b)$ and does not name the parameterisation.

The synthetic path uses Euler–Maruyama with time step 10^{-3} on $[0, 40]$ and is thinned by 2000. Twenty prey observations are kept, at times $2, 4, \ldots, 40$. The initial condition is treated as known. Inside each filter the transition between observations is ten Euler steps of length 0.2. A coordinate below 10^{-6} is replaced by 10^{-6}. The source paper does not state a boundary rule.

Three filters are run on this one series, with four algorithm replicates. The plain nested filter uses $\tilde{R} = H\hat{\mu}_{t|t-1}$ and targets $\hat{\pi}_{\text{enkf}}$. The Rao-Blackwellised filter multiplies that variance by 2, evaluates (3), and targets $\tilde{\pi}_{\text{enkf}}$. Bootstrap SMC2 uses the Euler–Maruyama transition

Table 1: Lotka–Volterra comparison. Mean, over four algorithm replicates, of the terminal weighted posterior mean of $\log \theta_j$, with the replicate standard deviation in parentheses. Draws are the mean number of Euler steps.

Filter	$\log \theta_1$	$\log \theta_2$	$\log \theta_3$	ESS	Draws
Plain	-0.714 (0.122)	-6.035 (0.104)	-1.276 (0.101)	33.3	448500
Rao-Blackwellised	-0.852 (0.084)	-5.971 (0.159)	-1.196 (0.156)	42.6	476100
SMC2	-0.837 (0.064)	-5.977 (0.138)	-1.241 (0.161)	43.5	905700

as its proposal and the observation density (27) as its state weight. For that discretisation its parameter target is π_t. Each filter uses $M = 60$ and $\gamma = 0.4$. The ensemble size is fixed at $N = 12$ for the first two filters. SMC2 uses 24 state particles. There is no delayed acceptance and no growth of N. The rejuvenation proposal is one random walk, with $\zeta^2 = 2.56^2/3$. Section 5.2 of Golightly et al. (2026) used $M = 1000$, an ensemble that starts at 20 and then grows, 50 replicates, delayed acceptance with three neighbours, and a particle Markov chain Monte Carlo gold standard of 10^6 iterations. None of those settings is used here. SMC2 is the reference for π_t, and it is a noisy one at this state-particle count.

The reported summary is the terminal weighted mean of $\log \theta_j$. Table 1 gives the average of that summary over the four replicates, with the replicate standard deviation in parentheses.

The pre-registered comparison is the plain-filter mean minus the SMC2 mean, judged against the standard error of that difference. For $\log \theta_2$ the difference is -0.0586, with standard error 0.0865. For $\log \theta_3$ the difference is -0.0354, with standard error 0.0953. Both differences are negative, so the point estimates lie in the direction reported by Golightly et al. (2026), and neither absolute difference exceeds its standard error. The shift is unresolved. It is not reported as appearing, and it is not reported as failing to appear. The Rao-Blackwellised mean is closer to the SMC2 mean on $\log \theta_2$, where the absolute gap is 0.0055 against 0.0586, and not on $\log \theta_3$, where the absolute gap is 0.0446 against 0.0354. The reduction on both coordinates, which the rule asked for, does not hold.

For $\log \theta_1$, which had no pass mark, the plain-filter mean exceeds the SMC2 mean by 0.1227, with standard error 0.0688. The Rao-Blackwellised gap on that coordinate is 0.0152. The mean number of proposals is 405 in each filter, and the mean numbers of acceptances are 79.25, 76.25 and 83.25. The largest fraction of Euler draws that met the floor is 0.008064, below the limit 0.01 fixed in the design. The draw counts in Table 1 do not include the observation-density evaluations inside (3), so they understate the cost of the Rao-Blackwellised weight relative to the plain filter.

5 Discussion

We have separated two errors that a nested ensemble Kalman filter can make, and we have compared the three targets those errors induce on a nonlinear observation. The plain filter targets $\hat{\pi}_{\mathrm{enkf}}$. A nearest-neighbour incremental weight targets π_{knn}. The Rao-Blackwellised weight puts the true observation density back and targets $\tilde{\pi}_{\mathrm{enkf}}$, which is still not the exact posterior. Bootstrap SMC2 is the scheme, among the three we ran, that targets π_t.

Lemma 1 is the linear Gaussian case of the Rao-Blackwellised weight. On the Ornstein–Uhlenbeck pilot the two log increments agreed exactly, so the correction is not worth its cost when the filter is already using the true observation density. The same pilot's surrogate diagnostic stayed inside the threshold fixed in advance: the mean of the median absolute log differences was 0.277, against a threshold of 1. A bound on the distance between π_{knn} and $\hat{\pi}_{\mathrm{enkf}}$ was not obtained. A uniform bound on the log-likelihood difference would control the total variation distance, and that hypothesis is not available for a likelihood that is redrawn with the ensemble.

On the Lotka–Volterra series the plain filter's means of $\log \theta_2$ and $\log \theta_3$ lie below the SMC2

means, by 0.0586 and 0.0354. The standard errors are 0.0865 and 0.0953, and the pre-registered rule therefore leaves the shift unresolved. The Rao-Blackwellised filter is closer to SMC2 on $\log \theta_2$ and not on $\log \theta_3$. Golightly et al. (2026) report a downward shift for the plain filter, reduced by the Rao-Blackwellised weight, at $M = 1000$ and against a long particle Markov chain Monte Carlo run. Our comparison uses $M = 60$, four replicates and SMC2 itself as the reference, so it can show the shift and it can miss it. Here it does not resolve it. The result does not confirm that ordering, and it does not contradict it.

The limits of the comparison are the ones the design named. There is one series. The state-particle count in SMC2 is fixed at 24, below the terminal count in Section 5.2 of Golightly et al. (2026), and a particle Markov chain Monte Carlo gold standard was not run. The standard errors are estimated from four replicates. A small fraction of Euler draws met the positive floor. The Euler counts omit the cost of the density ratio in the Rao-Blackwellised weight. The Gamma priors use a shape-rate reading that the source paper does not state. These are reasons to treat the unresolved verdict as a statement about this run.

An exact neighbour tree, a kernel switch that depends only on time, and a finite-difference proposal were part of the earlier pilot. They are not recommended here. The finite-difference proposal leaves $\hat{\pi}_{\text{enkf}}$ invariant and did not improve effective sample size per transition draw. What remains open is a bound on π_{knn} that can be checked, and a repeat of the Lotka–Volterra comparison at the particle counts of Section 5.2. We do not add an epidemic series. Temfack and Wyse (2026) already study a nested ensemble scheme for incidence.

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Data availability

All series used in this paper were simulated from the models in Section 4. The seeds and the numerical summaries are in the results files cited there.

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